

RESEARCH OVERVIEW

My research sits at the intersection of Computer Vision and Agentic AI, focusing on how models perceive, reason about, and act within visual and digital environments. This includes work on inverse graphics, procedural content generation, and computer-use agents that translate visual and expert knowledge into executable actions. I am eager to apply this focus to cutting-edge research in my Master's program.

EDUCATION

University of California, San Diego, CA, USA

Sep 2025 – Present

MS in Computer Science

- Concentration: Computer Vision & Graphics, ML & AI
- **GPA:** 3.95/4
- Advisor: [Manmohan Chandraker](#)

Indian Institute of Technology Gandhinagar, Gujarat, India

Nov 2021 – May 2025

B.Tech in Computer Science & Engineering

- **GPA:** 3.96/4 (Dean's List)
- Advisor: [Shanmuganathan Raman](#)

WORK EXPERIENCE

Graduate Researcher, UC San Diego – Center for Visual Computing, CA, USA

Sept 2025 – Present

BlendCUA: Self-Evolving Computer-Use Harness for Blender (first author)

- Built BlendCUA, a computer-use agent that turns natural-language briefs and reference images into Blender assets across modelling, materials, physics simulation and animation, and that improves its own harness (tools, skills, plan checks) from experience with the model kept frozen.
- Proposed learning from the defects in the agent's finished work instead of from successful trajectories, and implemented the loop end to end: an independent checker agent, a curator agent that attributes each defect to the plan, review, execution or reference views, and a rebuild gate that keeps a new skill or tool only if it fixes the defect.
- Outperformed existing harnesses such as Claude Code and Codex on held-out tasks; with most learned tools and skills reused on later tasks ([Video](#)).
- **Submission:** *BlendCUA: Self-Evolving Computer-Use Harness for Blender Use*, first author, under review at ICLR 2027.
- Advisors: [Manmohan Chandraker](#) (Prof., UCSD), [Kunal Gupta](#) (PhD, UCSD)

Procedural Material Generation from Expert Workflows (ECCV 2026)

- Researched agentic LLM systems that learn from expert video demonstrations and reason over expert workflows to produce executable outputs, applied to procedural materials in Blender.
- Knowledge extraction: built automated pipelines that turn Blender tutorial videos into structured procedural knowledge using video understanding and LLM-based reasoning.
- Generation: helped develop retrieval-time reasoning agents that use this knowledge to plan expert-style material creation processes and compile them into executable Blender node graphs.
- Supported expert and large-scale user studies showing improved workflow quality over graph-only generation baselines; paper (*Reflecting Process Expertise in Procedural Material Generation*) accepted at ECCV 2026.
- Advisors: [Manmohan Chandraker](#) (Prof., UCSD), [Kunal Gupta](#) (PhD, UCSD)

Co-founder, FluenTalk, India

Dec 2024 – Jun 2025

- Built a voice-based AI assistant web app for improving spoken English, tested with 200+ students ([Website](#), [Blog](#)).
- Developed a production web app integrating OpenAI APIs for real-time voice interaction, grammar correction, and fluency feedback across practice modes such as debates and passage reading.

- Architected the full-stack system: Django backend, React frontend, Firebase authentication, CockroachDB database.
- Designed ML-driven feedback mechanisms improving user engagement and measurable learning outcomes.

ML Intern, Indian Institute of Technology Gandhinagar – CVIG Lab, India

Sep 2024 – Jun 2025

- Worked on inverse modeling for outdoor scenes using single images.
- Used Gaussian Splatting and NeRF models to represent outdoor scenes and perform reconstruction/re-rendering on them.
- Advisors: [Shanmuganathan Raman](#) (Prof., IITGn), [Ashish Tiwari](#) (PhD, IITGn)

Research Intern, Purdue University – IDEAS Lab, West Lafayette, USA

May 2024 – Jul 2024

- Worked on operator learning (Neural Operators) to build surrogates for simulating fluids (Navier–Stokes equations).
- Worked on human pair motion generation using Generative Neural Operators.
- Advisors: [Aniket Bera](#) (Associate Prof., Purdue University), [Hrishikesh Vishwanath](#) (PhD, Purdue University)

GSoC Contributor, Google Summer of Code [Open Astronomy, Stingray]

May 2023 – Aug 2023

- Developed a Gaussian Processes inference tool that calculates the evidence of a QPO (Quasi-Periodic Oscillation) signal in astronomical time series from violent stellar events like GRBs and magnetar/solar flares.
- Made [demonstration and tutorial notebooks](#) elaborating on Bayesian inference and the project tool.
- Used various Python libraries: JAX, TinyGP, JAXNS, TensorFlow Probability.
- Open-source project funded under the Google Summer of Code program. Pull Request [\[webpage\]](#), Final Report [\[webpage\]](#)
- Advisors: [Daniela Huppenkothen](#) (University of Amsterdam), [Matteo Bachetti](#) (National Institute for Astrophysics, Italy)

PUBLICATIONS

1. **BlendCUA: Self-Evolving Computer-Use Harness for Blender Use**, under review at the International Conference on Learning Representations (ICLR) 2027 [\[video\]](#)
Gaurav Joshi, Kunal Gupta, Manmohan Chandraker
2. **Reflecting Process Expertise in Procedural Material Generation**, accepted at the European Conference on Computer Vision (ECCV) 2026 [\[webpage\]](#)
3. **Mitigating Simplicity Bias in Neural Networks: A Feature Sieve Modification, Regularization, and Self-Supervised Augmentation Approach** (SCSL @ ICLR 2025) [\[pdf\]](#) [\[webpage\]](#)
Gaurav Joshi, Parth Narendra Shah, Rachit Verma
4. **Stingray 2: A fast and modern Python library for spectral timing** (JOSS, October 2024) [\[pdf\]](#) [\[webpage\]](#)
Matteo Bachetti, Daniela Huppenkothen, Abigail Stevens, John Swinbank, Gaurav Joshi, et al.

TECHNICAL SKILLS

- **Research Interests:** Probabilistic/Bayesian Machine Learning, Astronomy, Mathematics, Deep Learning, High-Performance Computing, Numerical Analysis
- **Programming Languages:** Python, JAX, C, C++, Julia, Java
- **Other Skills:** GitHub, Jupyter Notebooks, Google Colab, Unix, Report Writing (LaTeX, Markdown, Jekyll), HTML

SCHOLASTIC ACHIEVEMENTS

2021–25	Academic Excellence Award / Dean's List , Undergraduate at IIT Gandhinagar	India
2024	Research Intern Fellowship , Purdue University, West Lafayette, Indiana	USA
2021–25	Erach and Meheroo Mehta Merit Scholarship , IIT Gandhinagar	India
2021	Ranked in the top 0.4 percentile , IIT-JEE (Advanced)	India
2019	National Fellowship , KVPY, Government of India	India
2019	National Scholarship , National Talent Search Examination, Government of India	India

KEY PROJECTS

PDE Solving using Physics-Informed Neural Networks [Self Project]

Oct 2023 – Dec 2023

- Implemented PINNs to solve Laplace equations for various Dirichlet conditions.
- Comparative evaluation with analytic and mesh methods.
- Tools used: TensorFlow, LaTeX
- Final Report [\[webpage\]](#)

Computational Statistics eBook [Jupyter Book]

Jul 2023 – Oct 2023

- Made an online book on computational statistics explaining the math behind important Bayesian data analysis and parameter estimation methods, along with code implementing those methods.
- Covered topics like random variable generation, Monte Carlo integration, MCMC methods (MH, Hamiltonian MC, Gibbs sampling), Gaussian Processes, information theory, Sequential Monte Carlo, graphical models, etc.
- Final Report [\[webpage\]](#)

Effective Sensor Placement using Neural Processes

Jul 2023 – Aug 2023

- Made an ML pipeline that uses Conditional Neural Processes for effective pollution sensor placement.
- Used CAMx simulated pollution data to predict future pollution levels and place sensors based on the Mutual Information criterion.
- Compared the quality of fit across models like Neural Processes and Gaussian Processes.
- Tools used: PyTorch, JAX, DeepSensor
- Mentor: [Prof. Nipun Batra](#)

Scientific Computing eBook [Julia eBook]

Mar 2023 – May 2023

- Made an online book on scientific computing with Julia.
- Focuses on the Julia language and Scientific Machine Learning.
- Final Report [\[webpage\]](#)

KEY COURSES UNDERTAKEN

UC San Diego, USA: Computer Vision I, Advanced Computer Vision, Advanced Image Synthesis, Selected Topics in Graphics, Deep Learning Systems, Machine Learning Systems, Data Systems for ML, Principles of Software Engineering.

IIT Gandhinagar, India: Deep Learning, AI & Machine Learning, Probabilistic ML, Natural Language Processing, Data Structures and Algorithms, Advanced Data Structures, Operating Systems, Computer Vision, Computer Graphics, Data Science, Computer Networks, Numerical Analysis & Computing, Advanced Mathematics, Computer Organization and Architecture, Discrete Mathematics.

Purdue University, USA: Algorithms, Computer Graphics, AI & Machine Learning, Computer Vision

ADDITIONAL INFORMATION

- GitHub: github.com/Gaurav17Joshi
- Blog: gaurav17joshi.github.io/contrast
- LinkedIn: [LinkedIn Profile](#)

REFERENCES

- [Manmohan Chandraker](#), Professor, Visual Computing Center and Contextual Robotics Institute, UCSD, USA
- [Shanmuganathan Raman](#), Jibaben Patel Chair Professor, Computer Science & Engineering, IITGn, India
- [Aniket Bera](#), Director of [IDEAS Lab](#), Associate Professor of Computer Science, Purdue University, USA
- [Daniela Huppenkothen](#), Staff Scientist, SRON Netherlands Institute for Space Research, Netherlands
- [Anirban Dasgupta](#), N Rama Rao Chair Professor, Computer Science & Engineering, IITGn, India